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Assignment No: 18

Problem Statement: Write a C program that accepts a string from the user and performs the following string operations: i) Calculate length of string ii) String reversal iii) Equality check for two strings iv) Check palindrome v) Check substring.

Objective:

- To understand basic string operations in C
- To learn how to manipulate strings using loops.
- To perform comparison, reversal and substring checking.
- To check whether a string is a palindrome.

Theory: In C programming, a string is a sequence of characters terminated by a null character. Strings are commonly stored in character arrays. Various operations can be performed on strings such as calculating their length, reversing the characters, comparing two strings, checking whether a string is a palindrome and finding a substring within another string.

Platform: Linux

Input: A string & another string for comparision & substring checking.

Output: Displays string length, reversed string, equality result, pallindrome result, & substring

check result.

Sample Input: Enter a string: madam
Enter another string for equality check: mad.

Sample Output: Length of string = 5
Reversed string = madam.
The string is a Palindrome
Strings are not equal.
Substring found in main string.

Conclusions: Thus, the C program successfully performs different ~~usi~~ string operations such as length calculation, reversal, equality, check, palindrome check, and substring detection.

FAQs:

1. What ~~a~~ is a string in C programming?
A In C programming, a string is a sequence of characters stored in an array and terminated by a special character called the null character

2. Which header file is required for string functions?
A In C programming, the header file required for string functions is:
#include <string.h>

3. What function is used to find the length of a string?
A. In C programming, the function used to find the length of a string is: strlen()

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a) What is a palindrome string?

A: A palindrome string is a string that reads the same forward and backward.
A string is called a palindrome if it remains the same when its characters are reversed

b) Which function is used to check a substring in C?

A: In C programming, the function used to check a substring is a string: strstr()